

Machine Learning-Based Classification of Heart Disease Risk for Insurance Decision Support Systems

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Abstract

This paper focuses on the classification of heart disease risk using machine learning methods to support medical prediction systems, with potential applications in insurance risk assessment. The study evaluates five individual classification algorithms, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (K-NN). These models are further combined using ensemble techniques such as Soft Voting, Hard Voting, and Stacking to enhance overall predictive accuracy. Experiments conducted on a Kaggle heart disease dataset consisting of 1,025 patient records and 13 predictive features show that individual models like SVM, Random Forest, and Logistic Regression yield high performance. However, the best results are achieved through ensemble methods, with Soft Voting obtaining the highest accuracy of 88.78%, followed closely by Stacking at 88.54%. Meanwhile, SVM, the best-performing individual model, achieved an accuracy of 88.41%. These findings demonstrate that ensemble approaches, particularly Soft Voting, offer superior predictive performance. This study supports the adoption of ensemble-based classifiers for effective and measurable heart disease risk prediction, contributing to the development of decision support systems in both clinical and insurance domains.

Keywords: Heart disease prediction, Machine learning, Ensemble method, Insurance risk, Classification model

1 Introduction

Heart disease continues to be one of the leading causes of death both globally and in Indonesia. As a vital organ, the heart is responsible for pumping blood, delivering oxygen, and distributing essential nutrients throughout the body. Any impairment in its function can significantly disrupt other organs and overall health [Hasan and Abdulazeez, 2024]. Global health statistics report that cardiovascular diseases account for millions of deaths each year and are projected to increase steadily in the coming years [Cahyani et al., 2024, Hasan and Abdulazeez, 2024]. In Indonesia, the rising prevalence of heart disease presents a serious public health concern while also creating economic challenges, particularly for the insurance sector. This underscores the growing need for proactive measures in prevention and early detection of cardiovascular conditions.

As defined by Kazi et al. [Kazi et al., 2024] in the United States, one in three adults received care for a cardiovascular risk factor or condition in 2020. Alarming, the annual inflation-adjusted (2022 USD) healthcare costs associated with cardiovascular risk factors are projected to triple—from \$400 billion in 2020 to \$1.344 trillion by 2050. Similarly, the costs for treating cardiovascular conditions are expected to almost quadruple, rising from \$393 billion to \$1.49

trillion, while productivity losses are projected to increase by 54%, from \$234 billion to \$361 billion. Stroke is projected to account for the largest absolute increase in costs. These trends highlight the global economic burden of cardiovascular disease and the urgent need for efficient, data-driven risk prediction models.

For insurance companies, early identification of individuals at high risk of heart disease is critical to accurately assess and manage risk exposure. Effective risk prediction enables insurers to implement appropriate underwriting policies, adjust premiums, and develop targeted preventive interventions, ultimately reducing potential financial losses [Benhar et al., 2020]. Therefore, leveraging data-driven approaches to predict heart disease risk has become essential in improving decision-making processes within the insurance industry.

Machine learning techniques, particularly classification algorithms, have demonstrated strong capabilities in processing complex medical datasets and generating timely, precise predictions [Hasan and Abdulazeez, 2024]. Among these, ensemble methods such as voting classifiers and stacking—which combine multiple individual models like logistic regression, k-nearest neighbor (KNN), and support vector machine (SVM)—offer enhanced predictive accuracy by aggregating decisions through majority voting or probabilistic averaging [Amaliana et al., 2024].

Given that the datasets used are generally well-balanced, reliance on oversampling methods like Synthetic Minority Oversampling Technique (SMOTE) can be minimized, allowing for more straightforward and effective modeling. The application of voting classifiers thus represents a promising approach to build robust predictive models tailored to the local context.

This study focuses on ensemble voting-based classification model to predict heart disease risk, aimed at supporting insurance companies in Indonesia with more accurate risk stratification. The expected outcome is to improve risk management strategies and facilitate better service provision to decision maker in insurance company through early and reliable detection of cardiovascular risk.

2 Related Work

Prior work includes several studies on heart disease prediction using machine learning and deep learning techniques, but they often lack direct applicability to insurance risk assessment in Indonesia. For example, [Perumal et al., 2018] applied classical classifiers such as logistic regression and support vector machines on the Cleveland dataset, achieving good accuracy but without considering ensemble methods that improve robustness. Similarly, [Gupta et al., 2019] used random forests to enhance prediction accuracy, yet their approach involved complex feature engineering not always feasible with real-world insurance data.

Regarding data preprocessing, [Benhar et al., 2020] provided a systematic review highlighting the importance of data cleaning and reduction, but did not focus on well-balanced datasets typical in Indonesian insurance. Deep learning models like convolutional neural networks have shown impressive results in detecting arrhythmias and coronary artery disease [Hannun et al., 2019, Zreik et al., 2019], yet they require large datasets and suffer from interpretability issues, limiting their practical use in insurance underwriting.

Unlike these prior works, our approach leverages an ensemble voting classifier combining stacking, logistic regression, K-NN, random forest, decision tree, and SVM tailored for well-balanced datasets, minimizing the need for oversampling techniques. This method offers a simpler, interpretable, and effective solution to support early heart disease risk prediction specifically designed to improve decision-making in insurance risk management.

3 Methodology

3.1 Data Collection

The dataset used in this study was obtained from the Kaggle platform, titled *Heart Disease Dataset*, published by user johnsmith88¹. This dataset is open-access and has been

widely utilized in machine learning-based heart disease prediction studies.

The data file is in CSV format under the name `heart.csv`, consisting of 1,025 patient records and 13 predictor features, along with 1 binary target variable indicating the presence of heart disease. The features represent a combination of demographic and clinical examination data such as age, blood pressure, cholesterol level, and electrocardiogram results. The target variable is encoded as 0 (no heart disease) and 1 (presence of heart disease). Table 1 presents the description of the dataset features, data types, and medical interpretations.

Table 1: Feature description of the Cleveland Heart Disease dataset

Feature	Type	Description
age	Integer	Age of the patient in years
sex	Categorical	Biological sex (0 = female, 1 = male)
cp	Categorical	Chest pain type (0: typical angina, 1: atypical angina, 2: non-anginal pain, 3: asymptomatic)
trestbps	Integer	Resting blood pressure (in mm Hg)
chol	Integer	Serum cholesterol level (in mg/dl)
fb	Categorical	Fasting blood sugar > 120 mg/dl (0 = false, 1 = true)
restecg	Categorical	Resting electrocardiographic results (0: normal, 1: ST-T wave abnormality, 2: left ventricular hypertrophy)
thalach	Integer	Maximum heart rate achieved
exang	Categorical	Exercise-induced angina (0 = no, 1 = yes)
oldpeak	Float	ST depression induced by exercise relative to rest
slope	Categorical	Slope of the peak exercise ST segment (0: upsloping, 1: flat, 2: downsloping)
ca	Integer	Number of major vessels colored by fluoroscopy (range: 0–3)
thal	Categorical	Thalassemia test result (0: normal, 1: fixed defect, 2: reversible defect)
target	Binary	Presence of heart disease (0 = no, 1 = yes)

¹<https://www.kaggle.com/datasets/johnsmith88/heart-disease-dataset>

3.2 Model Architecture

This study implements five classification algorithms as base models: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbor (KNN), which represent various approaches in supervised learning. Subsequently, three additional approaches were employed, namely Voting Classifier and Stacking Classifier as ensemble methods, and a comprehensive evaluation framework using cross-validation. This architecture is designed to compare model performance while optimizing predictive results through ensemble integration. The following subsections describe each method in detail.

3.2.1 Logistic Regression

Logistic Regression is a binary classification algorithm that models the relationship between input features and the probability of a target class using the sigmoid function. The model creates a linear combination of features and coefficients and transforms it into a probability with the following function:

$$P(y = 1 | X) = \frac{1}{1 + e^{-z}}, \quad (1)$$

$$\text{where } z = \beta_0 + \sum_{i=1}^n \beta_i X_i$$

where X_i represents the input features, β_i denotes the coefficient (weight) corresponding to each feature, and β_0 is the intercept (bias) of the model. The output of this function is interpreted as the predicted probability of the positive class.

Logistic Regression is interpretable and computationally efficient, and is commonly used as a baseline in medical classification tasks. It is suitable when the relationship between features and the log odds of the outcome is linear [Chang et al., 2022, Ingole et al., 2024].

3.2.2 Decision Tree

A Decision Tree builds a tree-based structure to perform classification decisions. Each internal node represents a condition on a feature, and the splits are determined using metrics such as Gini Impurity or Information Gain. The classification result is decided by the majority class at the leaf node:

$$Gini = 1 - \sum_{i=1}^k p_i^2 \quad (2)$$

where p_i is the proportion of samples belonging to class i in a given node, and k is the total number of classes. A smaller Gini value indicates that the node is more pure (homogeneous).

Decision Trees are highly interpretable, but they are prone to overfitting if not properly pruned [Mandal et al., 2025].

3.2.3 Random Forest

Random Forest is an ensemble method that combines multiple decision trees built using bootstrap sampling and random feature selection. The final prediction is made through a majority voting scheme across all trees:

$$\hat{y} = \text{majority vote}\{h_1(x), h_2(x), \dots, h_T(x)\} \quad (3)$$

where $h_t(x)$ denotes the prediction of the t -th decision tree for the input x , and T is the total number of trees in the ensemble. The final output $\hat{y}$ is determined by the majority vote across all individual tree predictions.

Random Forest is known for its robustness against overfitting and its ability to capture nonlinear relationships between features [Chang et al., 2022, Ingole et al., 2024].

3.2.4 Support Vector Machine

Support Vector Machine (SVM) constructs an optimal hyperplane to separate classes with maximum margin. For non-linearly separable data, SVM applies kernel functions such as the Radial Basis Function (RBF) to project the data into a higher-dimensional space:

$$f(x) = \sum_{i=1}^n \alpha_i y_i K(x_i, x) + b \quad (4)$$

where x_i is a support vector, y_i is the class label associated with x_i , α_i is the Lagrange multiplier determined during training, $K(x_i, x)$ is the kernel function that measures the similarity between x_i and the input x , and b is the bias term. The output $f(x)$ is used to determine the class of the input x based on the sign of the predicted value.

SVM performs well on high-dimensional datasets and generally produces models with good generalization [Mandal et al., 2025].

3.2.5 K-Nearest Neighbor

K-Nearest Neighbor (KNN) is an instance-based learning algorithm that classifies new samples based on the majority label among their k closest neighbors, typically determined by Euclidean distance:

$$d(x, x') = \sqrt{\sum_{i=1}^n (x_i - x'_i)^2} \quad (5)$$

where x and x' represent the feature vectors of two samples, and n is the number of features. The Euclidean distance measures how close the two samples are in the feature space and is used to identify the nearest neighbors of a test instance.

KNN is simple and effective but sensitive to the choice of k and the scaling of features [Mandal et al., 2025].

3.2.6 Ensemble Voting Classifier

Voting Classifier is an ensemble technique that combines the predictions of multiple base models. There are two main types of voting:

- **Hard Voting:** the class with the majority of votes is chosen.
- **Soft Voting:** the class with the highest average predicted probability is chosen.

This method enhances prediction stability and accuracy by leveraging the strengths of diverse models with complementary error patterns [Amaliana et al., 2024, Chang et al., 2022].

3.2.7 Ensemble Stacking Classifier

Stacking Classifier is a more advanced ensemble method that combines several base models and uses a meta-learner to learn the optimal combination of their predictions. The outputs of the base models serve as inputs to the meta-learner:

$$\hat{y} = g(h_1(x), h_2(x), \dots, h_n(x)) \quad (6)$$

where $h_1(x), h_2(x), \dots, h_n(x)$ are the prediction outputs of each base model for the input x , and $g(\cdot)$ is the function of the meta-learner that learns the optimal combination of these predictions to produce the final output $\hat{y}$.

Stacking often outperforms simple voting because it captures interactions between base model outputs [Prasetya et al., 2024].

3.2.8 Evaluation Method

Model performance is assessed using cross-validation and standard performance metrics, including accuracy, precision, and F1-score, to ensure reliable and interpretable results.

Cross-Validation. To mitigate overfitting and evaluate model consistency, k -fold cross-validation is applied, where the data is divided into k folds and the model is trained and tested k times. This study adopts 5-fold cross-validation, consistent with prior works such as Mandal et al. [2025] and Chang et al. [2022], which emphasize the importance of multi-metric evaluation and robustness testing in medical classification tasks.

Evaluation Metrics. Following the validation process, the model's predictive performance is measured using standard performance metrics commonly applied in medical machine learning:

- **Accuracy**, which calculates the proportion of correctly classified instances relative to the total number of observations:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

where TP is the number of true positives, TN is the number of true negatives, FP is the number of false positives, and FN is the number of false negatives.

- **Precision**, which quantifies the proportion of true positive predictions among all instances predicted as positive:

$$\text{Precision} = \frac{TP}{TP + FP} \quad (8)$$

where TP is the number of true positives, and FP is the number of false positives.

- **Recall (Sensitivity)**, which measures the proportion of actual positive cases that are correctly identified by the model:

$$\text{Recall} = \frac{TP}{TP + FN} \quad (9)$$

where TP is the number of true positives, and FN is the number of false negatives.

- **F1-score**, the harmonic mean of precision and recall, which provides a balanced measure particularly useful in datasets with imbalanced class distributions:

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

where *Precision* and *Recall* are the values as previously defined.

3.3 Experimental Setup

The experiments in this study were conducted using the Python programming language on the Google Colaboratory (Google Colab) platform. This cloud-based environment was chosen for its scalability and compatibility with open-source libraries such as `pandas`, `numpy`, and `scikit-learn`, which were used throughout the model training and evaluation pipeline.

The experimental workflow consists of six major steps:

1. **Data loading and exploration:** Loading the dataset from a CSV file, followed by checking for data types, class distribution, and missing values.
2. **Data preprocessing:** This includes handling missing values (if any), encoding categorical variables, and normalizing numerical features.
3. **Data splitting:** Dividing the data into training and testing sets in an 80:20 ratio using `train_test_split()`.
4. **Model training:** Training five classification models — Logistic Regression, Decision Tree, Random Forest, SVM, and KNN — using the `fit()` method.
5. **Ensemble methods:** Combining the best-performing models (Random Forest, SVM, Logistic Regression) into a Voting Classifier (hard and soft voting) and a Stacking Classifier with Gaussian Naive Bayes as meta-learner.

6. **Evaluation:** Computing accuracy, precision, recall, and F1-score using `classification_report` and `confusion_matrix`, and visualizing performance via `ConfusionMatrixDisplay`. A 5-fold cross-validation was performed specifically on the Stacking Classifier.

4 Results and Discussion

4.1 Exploratory Data Analysis

Before building the classification model, we will explore the data as a first step. Exploratory data analysis is important in heart disease prediction as it helps reveal relationships among features, understand data characteristics, and identify underlying patterns [Indrakumari et al., 2020].

In this case study, the data used consisted of 1,025 rows and 14 columns. To provide an initial overview of the data characteristics, a descriptive analysis was performed on each column, which includes the measure of centering and the distribution of values. The results of the descriptive analysis can be seen in the following table.

Table 2: Descriptive Statistics

Variable	Mean	Minimum Value	Maximum Value
age	54.44	29	77
trestbps	131.62	94	200
chol	246	126	564
thalach	149.12	71	202
oldpeak	1.07	0	6.2
ca	0.75	0	4

Table 2 shows the summary results of descriptive statistics of numerical variables. It can be seen that the mean age of the respondents was 54 years old and ranged from 29 to 77 years old, indicating that the majority of respondents fell into the middle-aged to elderly category. The average resting blood pressure (131.62 mmHG) and cholesterol (246 mg/dL) were above the normal threshold.

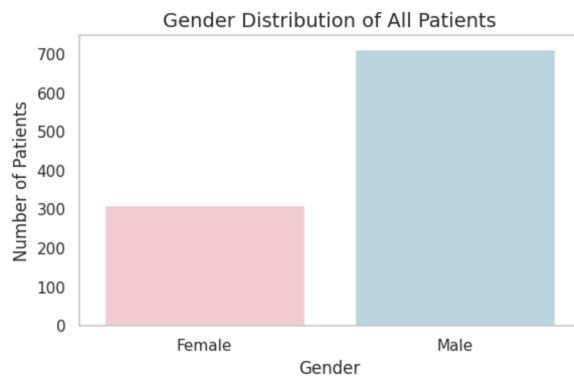


Figure 1: Target Class Distribution

Based on Figure 1, it can be seen that the majority of re-

spondents are male, which is around 700 people, while female respondents number around 300 people.

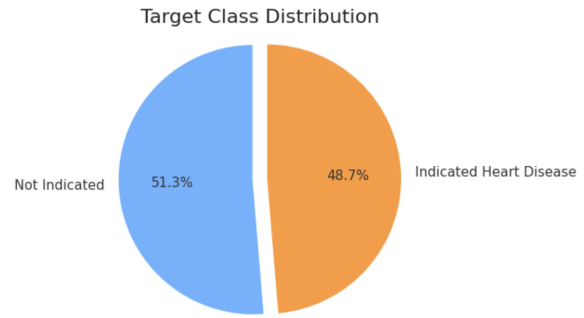


Figure 2: Target Class Distribution

Figure 2 illustrates the distribution by target group. It can be seen that out of 1,025 patients data, 51.3% of patients are not indicated to have heart disease, while the other 48.7% are indicated to have heart disease. This shows that the class distribution is quite balanced.

After looking at the target class distribution, the next step is to analyze the demographic characteristics of patients with heart disease. The following presents a visualization of the age and gender distribution of patients with heart disease to provide a more detailed picture of the patient group.

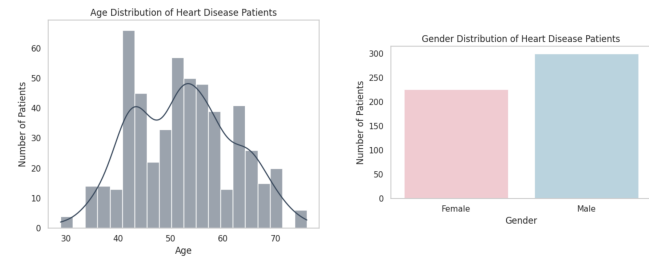


Figure 3: Age Distribution of Heart Disease Patients

Figure 4: Gender Distribution of Heart Disease Patients

Figure 3 shows that the number of patients with heart disease increases with age, with the highest peak in the 50s. However, the highest number of people with heart disease are in their 40s. This indicates that there are other influencing factors. Meanwhile, Figure 4 shows that patients who are indicated to have heart disease are dominated by men as many as around 300 people, while female patients number around 220 people.

Furthermore, a correlation analysis between numerical features is carried out to understand the relationship between variables that can affect the prediction of heart disease.

Figure 5 shows the correlation matrix between numerical variables and target variables. The feature with the highest positive correlation to the target is `thalach`, with a value of 0.42. This indicates that the higher the `thalach` (maximum heart rate) value, the higher the probability of heart disease. In contrast, the `oldpeak` feature has a negative correlation with the target of -0.44. This means that the higher the `oldpeak`

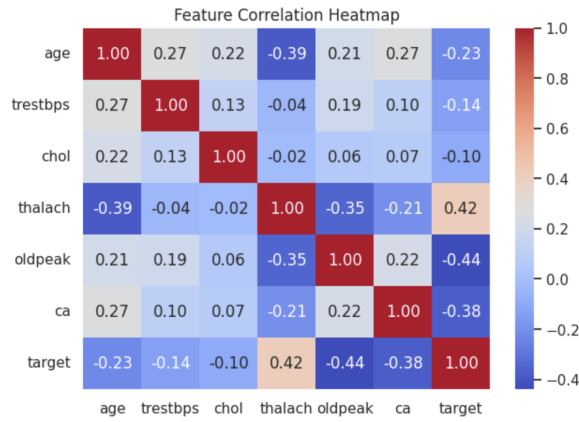


Figure 5: Feature Correlation Heatmap

value (ST segment depression), the lower the probability of heart disease.

4.2 Results

Before evaluating the performance of the model, it is necessary to know the class distribution of the target variable. This information is important because in the case of class imbalance, metrics such as accuracy can be misleading for example, a model that always predicts the majority class can still achieve high accuracy without really understanding the data pattern.

Here is the class distribution on the target variable.

Table 3: Class Distribution on Target

Class	Total
1	526
0	499

Table 3 shows the class distribution on the target. It can be seen that the target classes are relatively balanced, with 526 at-risk samples and 499 non-risk samples. Since the class distribution is quite balanced, the accuracy metric can be validly used as a basis for evaluation.

Initial testing of five machine learning algorithms namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbor (K-NN) was conducted by comparing prediction performance on test data using the confusion matrix (accuracy, precision, and F1-score) and accuracy metrics. However, to evaluate the performance in a more stable manner and prevent dependency on a single subset of data, cross-validation is used as the main evaluation method, with the average accuracy as the reference for selecting the best model.

Although the main evaluation is done using cross-validation, the evaluation results on the test data are still included as a supplement to provide an overview of the model's

Table 4: Accuracy with Cross-Validation

Model	Accuracy
SVM	0.8841
Random Forest	0.8768
Logistic Regression	0.8549
KNN	0.8537
Decision Tree	0.8220

performance when applied to real data. The following table presents a comparison of the accuracy, precision, and F1-score performance metrics based on the prediction results against the test data.

Table 5: Performance comparison

Model	Accuracy	Precision	F1-Score
Logistic Regression	0.80	0.80	0.79
Decision Tree	0.78	0.79	0.77
Random Forest	0.80	0.80	0.79
SVM	0.83	0.84	0.83
K-NN	0.80	0.81	0.80

Table 5 shows performance metrics. From the evaluation results on the test data shown in Table 5, it can be seen that the SVM model has the highest performance, followed by Random Forest and Logistic Regression, both in terms of accuracy and F1-score. However, since evaluation on test data is prone to overfitting and does not represent the performance of the model in general, the selection of the best model is still based on the evaluation results using cross-validation techniques.

To complement the quantitative analysis that has been carried out, the next step is to present a visualization of the model prediction results in the form of a confusion matrix. This visualization aims to provide a more detailed picture of the model's ability to classify data, especially in distinguishing between patients at risk and those not at risk of heart disease. By looking at the distribution of True Positive, True Negative, False Positive, and False Negative values, the evaluation of model performance becomes more comprehensive and informative.

Figure 6 shows the confusion matrix visualization of each classification model that has been tested, namely Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbor (K-NN). Each confusion matrix illustrates the number of correct and incorrect predictions based on test data. From this confusion matrix visualization, it can be seen that SVM has the best performance with the least number of prediction errors, both for positive and negative cases. However, Random Forest also shows excellent and stable performance.

The accuracy results of all classification models tested were compared to determine the best performing model (Table 5). The three models with the highest accuracy, namely Support Vector Machine (SVM), Random Forest, and Logistic Regression, were selected for use in the ensemble method.

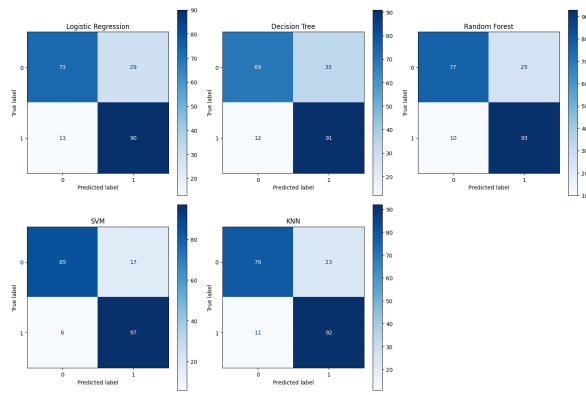


Figure 6: Architecture of the proposed model.

Therefore, an ensemble machine learning approach using voting classifiers, namely soft voting and hard voting, was used to combine predictions from several single classification models to obtain more accurate classification results. In addition, the Stacking Classifier method was used as an advanced ensemble approach to improve the overall performance of the model.

The comparison of the best performing models is then followed by a more thorough further evaluation, by comparing the accuracy of the three best base models with several ensemble models, namely the Voting Classifier (in the form of soft voting and hard voting) and the Stacking Classifier. This evaluation was conducted using a cross-validation approach to obtain a more stable performance picture.

Table 6: Accuracy with Cross-Validation

Model	Accuracy
SVM	0.8841
Random Forest	0.8768
Logistic Regression	0.8549
Soft Voting	0.8878
Hard Voting	0.8829
Stacking	0.8854

Table 6 shows the accuracy evaluation results of each model using the cross-validation method. These values are used as a basis for consideration in determining the best model for heart disease risk classification cases. Based on these results, the Soft Voting Ensemble model recorded the highest accuracy of 88.78%, followed by Stacking with 88.54%.

To complement the quantitative analysis of cross-validation, an evaluation of model performance was also conducted on test data using accuracy, precision, and F1-score metrics. This evaluation aims to see the consistency of the model's performance when applied to data that is not involved in the training process.

Table 7 shows the performance evaluation results of the ensemble models on the test data. It should be noted that the

Table 7: Ensemble Model Comparison

Model	Accuracy	Precision	F1 Score
Soft Voting	0.82	0.84	0.82
Hard Voting	0.82	0.83	0.82
Stacking	0.83	0.84	0.82

results on the test data may differ from the cross-validation results due to the different evaluation approaches. Evaluation on test data uses only a specific subset of data that may not fully represent the overall distribution of the data, making the results more susceptible to local biases or fluctuations. In contrast, cross-validation divides the data into multiple subsets and performs training and testing in turn, so the average result provides a more stable and generalizable performance estimate.

For example, the SVM model obtained an accuracy of 0.83 on the test data, but recorded a higher average accuracy of 0.8841 on the cross-validation evaluation. Similarly, the Soft Voting model recorded the highest accuracy of 0.8878 on cross-validation, but only achieved an accuracy of 0.82 when tested on test data. This difference shows the importance of using a comprehensive evaluation approach and not just relying on a single subset of data.

In general, ensemble methods tend to yield higher cross-validation accuracy than individual base models. This is because ensemble methods, such as Soft Voting or Stacking, are designed to combine the strengths of multiple models and reduce their individual weaknesses. By aggregating the predictions from several diverse base models, ensembles can better generalize across different data subsets and mitigate the risk of overfitting to specific patterns in the training data. As a result, they are more robust to variance and often achieve more consistent and improved performance during cross-validation. Moreover, ensemble models benefit from the "wisdom of the crowd" principle, where combined decisions from multiple models are statistically more accurate than any single model alone, especially when the base models contribute complementary decision boundaries.

Considering the stability and better overall performance based on the cross-validation results, the Soft Voting model was chosen as the best model. The advantage of this model lies in its ability to combine the predictive power of several base models, namely SVM, Random Forest, and Logistic Regression. The Soft Voting approach works by combining the prediction probabilities of the three models, then selecting the class with the highest probability as the final result.

Each base model has its own strengths: SVM excels at separating classes with maximum margin, Random Forest is effective in handling complex data and reducing the risk of overfitting through ensemble decision trees, while Logistic Regression offers simplicity, interpretability, and strong performance when the relationship between features and the target variable is approximately linear. This combination results in predictions that are more balanced and resilient to data variation, improving overall accuracy compared to individual

models.

4.3 Visualization Example

The architecture of the proposed model in this study consists of several main stages, namely data preprocessing, model building, model performance evaluation, and application of ensemble methods to improve prediction accuracy. All of these stages are explained as follows:

1. **Data Preprocessing:** This stage aims to prepare the data so that it can be used optimally by machine learning models. The preprocessing process includes:

- **Label Encoding** on the target variable to convert category labels into numeric values (0 and 1).
- **One-Hot Encoding** for categorical features such as sex, cp, fbs, restecg, exang, slope, thal. **Standardization** items on numeric features such as age, trestbps, chol, thalach, oldpeak, and ca use StandardScaler to scale the data uniformly.

2. **Model Building:** After preprocessing, a classification model is built using five machine learning algorithms, namely:

- Logistic Regression
- Decision Tree
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbor (K-NN)

3. **Model Evaluation:** All models are evaluated using a confusion matrix that compares the predicted and actual values of the test data. From this confusion matrix, evaluation values such as accuracy, precision, and F1-score are calculated to assess the performance of each model.

4. **Confusion Matrix Visualization:** The confusion matrix visualization of each model is used to provide a more detailed picture of the model's performance in classifying inpatients at risk and not at risk of heart disease.

5. **Cross-Validation:** To obtain a more stable picture of model performance that does not depend on a particular subset of test data, evaluation of the best models both single and ensemble models was carried out using the *5-fold cross-validation* technique. This approach is used consistently in this report as the main evaluation method in selecting the best model.

6. **Ensemble Learning:** The three models with the highest accuracy based on the evaluation using the cross-validation technique, namely Random Forest, SVM, and Logistic Regression, were selected as the base models in the ensemble approach. The ensemble approach applied includes:

- Soft Voting Classifier

- Hard Voting Classifier
- Stacking Classifier

This approach aims to combine the advantages of each model to produce more accurate and stable predictions.

7. **Final Model Selection:** All models, both single and ensemble, are then re-evaluated using five-fold cross-validation to determine the best model based on the highest average accuracy. This process ensures that the selected model has the most consistent and generalizable performance against new data.

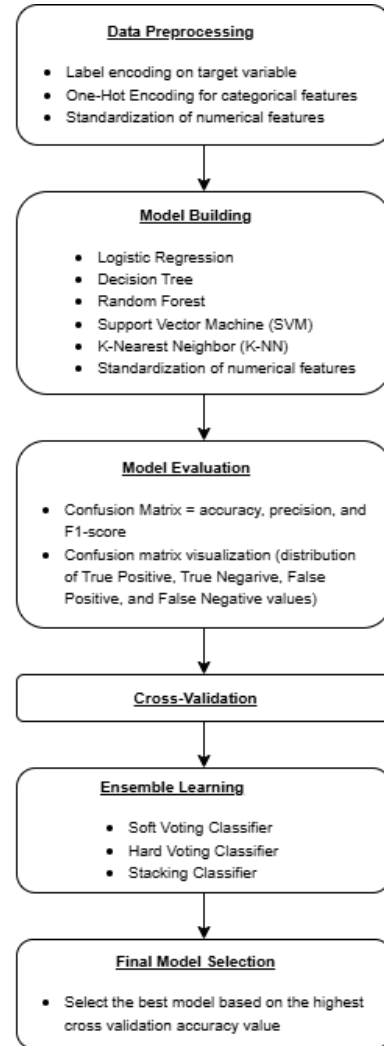


Figure 7: Architecture of the proposed model.

Figure 7 shows the architectural flow of the proposed model from the preprocessing stage to the final evaluation stage, which is expected to significantly improve the accuracy of heart disease prediction.

4.4 Limitations

The approach used in this study may not generalize well to populations beyond the original dataset's context. The data

utilized comes from a public Kaggle dataset, which primarily represents patients from outside Indonesia. Therefore, the patterns learned by the model may not fully reflect the clinical characteristics or risk factors commonly found in the Indonesian population. Furthermore, the dataset used may not capture the most recent developments in the medical field. In the ever-evolving medical context, having access to newer and locally relevant data is essential to improve the accuracy and validity of predictions.

For further development, the model can be tested on external datasets that are more locally representative to better align with the Indonesian population. Additionally, exploring alternative approaches such as deep learning or boosting methods could be pursued to enhance the accuracy and reliability of the predictions. With support from more up-to-date medical data, the resulting model is expected to be stronger and more dependable in real-world applications.

5 Conclusion and Future Work

This paper developed a classification model to predict heart disease risk by comparing the performance of several machine learning algorithms, namely Support Vector Machine (SVM), Random Forest, and Logistic Regression, as well as ensemble approaches such as Soft Voting, Hard Voting, and Stacking Based on the evaluation results, individual models such as SVM demonstrated high accuracy (88.41%), followed by Random Forest (87.68%) and Logistic Regression (80.85%).

However, ensemble methods provided better results. The Soft Voting model achieved the highest accuracy of 88.78%, outperforming all individual and other ensemble models. The Stacking model also showed competitive performance with an accuracy of 88.54%. The visualization of the confusion matrix supports these findings, showing that ensemble methods reduce prediction errors in both classes.

Therefore, ensemble approaches particularly Soft Voting have been proven effective in improving prediction accuracy and are recommended for implementation in decision support systems for clinical and insurance purposes. Future research will focus on validating the model using external datasets and enhancing interpretability to support practical deployment.

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